# Muneeb Abbasi

Principal AI Engineer — LLMs, RAG & agentic systems · 15+ years · ex-Microsoft

Kent, WA · abbasiamuneeb@outlook.com · amuneeb.github.io · linkedin.com/in/abdul-muneeb-abbasi · github.com/amuneeb

## Summary

Principal Software Engineer with 15+ years delivering enterprise-scale software, AI platforms, cloud-native systems, and workflow automation. Hands-on expertise in LLMs, RAG, AI agents, AWS, Azure, .NET Core, Python, React, microservices, and data platforms. Proven ability to lead technical discovery and solution architecture with customers, then own delivery end to end — from design through production adoption — with measurable business outcomes.

## Technical expertise

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| **AI & applied intelligence** | Generative AI, LLMs, RAG, AI agents, agentic workflows, Amazon Bedrock (Claude, Titan, Llama), AWS Knowledge Bases (RAG), Titan embeddings, Amazon SageMaker, AI guardrails, semantic & vector search, prompt engineering, OpenAI / Azure OpenAI, LangGraph, OCR, scikit-learn |
| **Cloud & infrastructure** | AWS (Lambda, Step Functions, EventBridge, EKS/ECS, API Gateway, Cognito, CloudFront, WAF, Route 53), Microsoft Azure, Azure Functions, Event Hub, Data Factory, Azure AI Services, Docker, Kubernetes, GitHub Actions, Azure DevOps, event-driven architecture, blue/green & canary deployments |
| **Software engineering** | Python, C#, .NET Core, TypeScript, JavaScript, Java, SQL, React, Angular, FastAPI, microservices, REST/GraphQL APIs |
| **Data & storage** | Amazon Aurora PostgreSQL, DynamoDB, Amazon Redshift, OpenSearch, ElastiCache (Redis), S3 data lake, SQL Server, Azure SQL, MongoDB, vector databases, data pipelines, telemetry analytics |

## Professional experience

### Principal AI Engineer — Global E-commerce & Retail

*Bellevue, WA · Feb 2025 – Present*

AI-Powered Inventory & Operations Platform — an enterprise ERP for a global e-commerce and retail business, built on AWS with an event-driven, cloud-native, AI-first architecture.

- Built the AI platform on Amazon Bedrock (Claude, Titan, Llama): RAG with AWS Knowledge Bases over an OpenSearch vector store and Aurora pgvector, Titan embeddings, and Amazon SageMaker models for demand forecasting, lead scoring, delivery ETA/risk prediction, and anomaly detection — reducing manual reconciliation by 90%
- Designed domain AI agents (quoting, procurement, warehouse, delivery exceptions, warranty, finance, executive insights) with agent orchestration, tool/function calling, and human-in-the-loop approval — protected by AI guardrails: content filtering, PII protection, prompt-injection protection, tool allow-lists, action approval, and audit logging
- Delivered event-driven microservices across 18+ business domains — EventBridge, SNS/SQS, Step Functions, Lambda, containers on EKS/ECS — with circuit breakers, rate limiting, idempotency, and a schema registry
- Data layer on Amazon Aurora PostgreSQL with read replicas, DynamoDB, Redshift, S3 data lake with Glacier archive, OpenSearch, and ElastiCache (Redis); applications in Python, FastAPI, and React behind API Gateway (REST/GraphQL) with Cognito, CloudFront, WAF, and Route 53
- Document intelligence with Amazon Textract and Comprehend; observability via CloudWatch, X-Ray, and CloudTrail, with blue/green and canary deployments

### Senior Software Engineer — Microsoft

*Redmond, WA · Oct 2014 – Jan 2025*

Led technical discovery, solution architecture, and delivery of cloud-native automation and AI-powered platforms for Microsoft's global network infrastructure, owning solutions from requirements through production while remaining a hands-on engineer.

**Network Infrastructure Copilot (NIC)**

- Built an enterprise AI copilot (LLMs, RAG, semantic search, telemetry) for natural language access to KPIs, incidents, and operational systems
- Increased engineering productivity by 35% and reduced incident investigation time across global infrastructure

**Enterprise AI knowledge & orchestration platform**

- Designed RAG and vector-search architecture unifying enterprise knowledge across engineering systems and documentation
- Reduced engineering research and decision-making time by 60%; improved onboarding and cross-team collaboration

**Network device automation system**

- Designed cloud-native microservices (.NET Core, Azure Functions, event-driven architecture) orchestrating global deployment workflows
- Reduced manual device scheduling by 60% with 99.9% platform availability

**Business process optimization platform**

- Automated enterprise workflows with orchestration, telemetry, operational intelligence, and AI-driven recommendations
- Improved business process efficiency by 40% and increased customer adoption

## Education

- Master of Business Information Systems (MBS) — University College Cork
- Bachelor of Engineering in Information Technology (B.Eng.) — Hamdard University